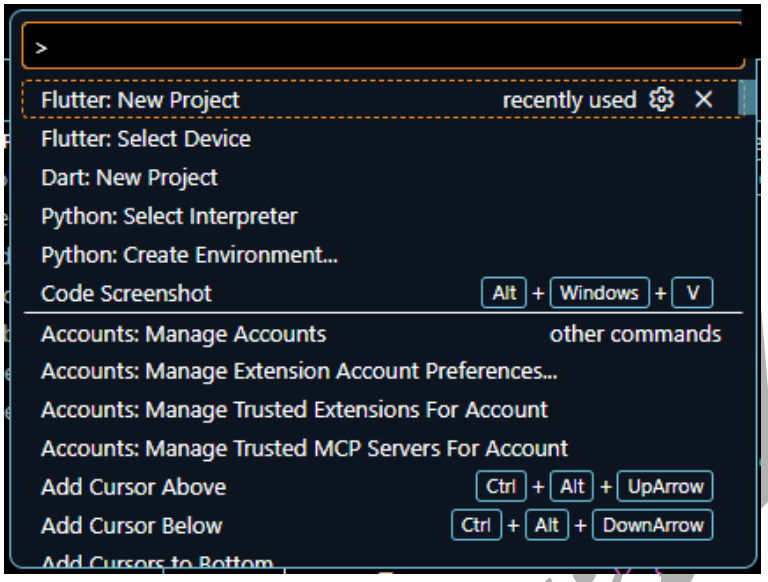
NAME : MARSYA ZULFA ZAHRANI
NIM : 244107060005
ABSENT : 11
CLASS : SIB 2G
OBJECT : MOBILE PROGRAMMING

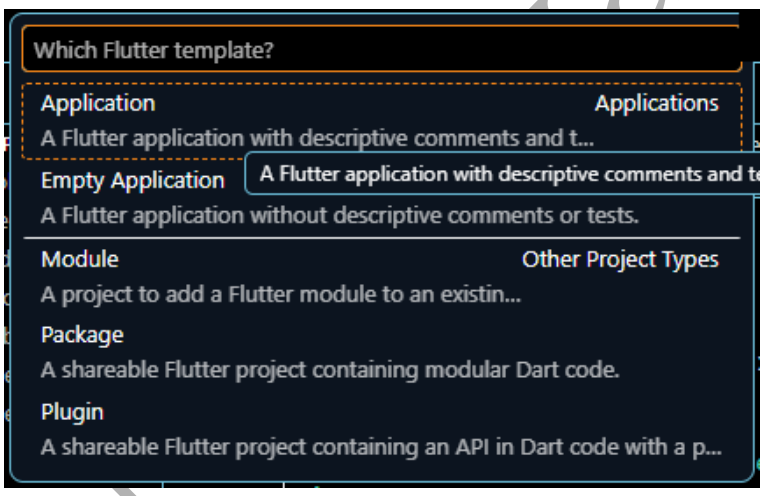
Jobsheet Flutter 1

Praktikum 1: Membuat Project Flutter Baru

Langkah 1:

Buka VS Code, lalu tekan tombol Ctrl + Shift + P maka akan tampil *Command Palette*, lalu ketik Flutter. Pilih New Application Project.





Langkah 2:

Kemudian buat folder sesuai style laporan praktikum yang Anda pilih. Disarankan pada folder dokumen atau desktop atau alamat folder lain yang tidak terlalu dalam atau panjang. Lalu pilih Select a folder to create the project in.

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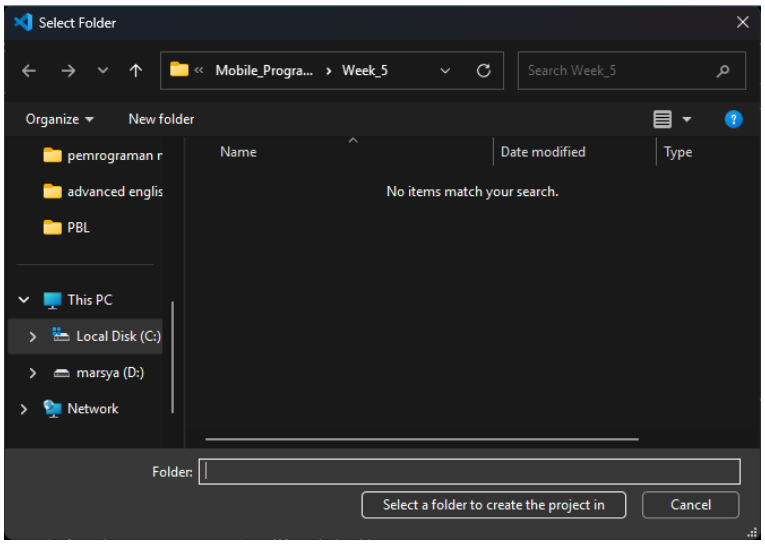
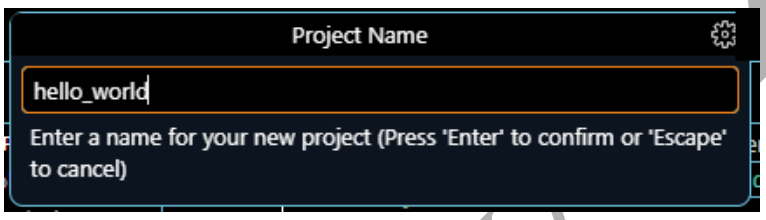


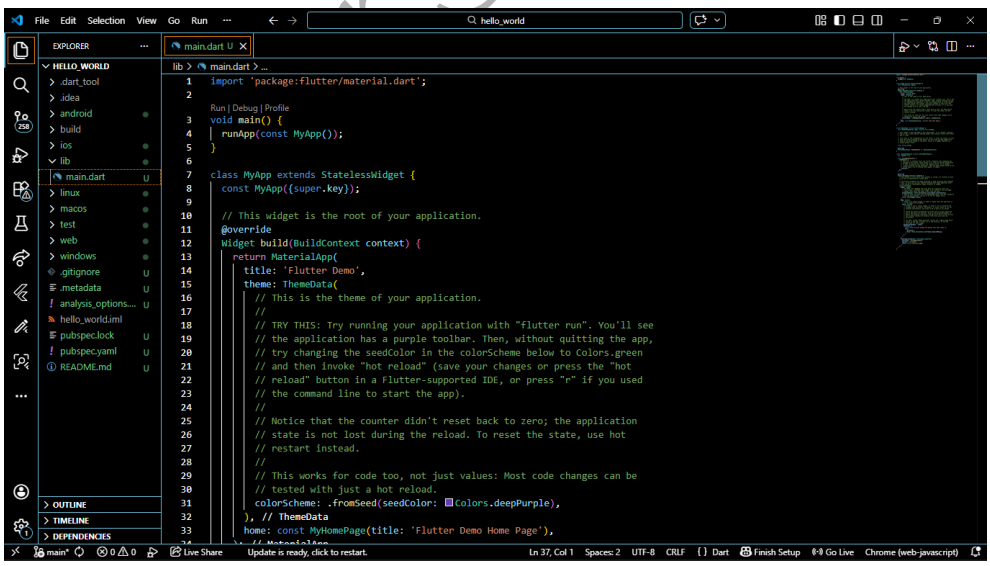
Langkah 3:

Buat nama project flutter hello_world seperti berikut, lalu tekan Enter. Tunggu hingga proses pembuatan project baru selesai.



Langkah 4:

Jika sudah selesai proses pembuatan project baru, pastikan tampilan seperti berikut.



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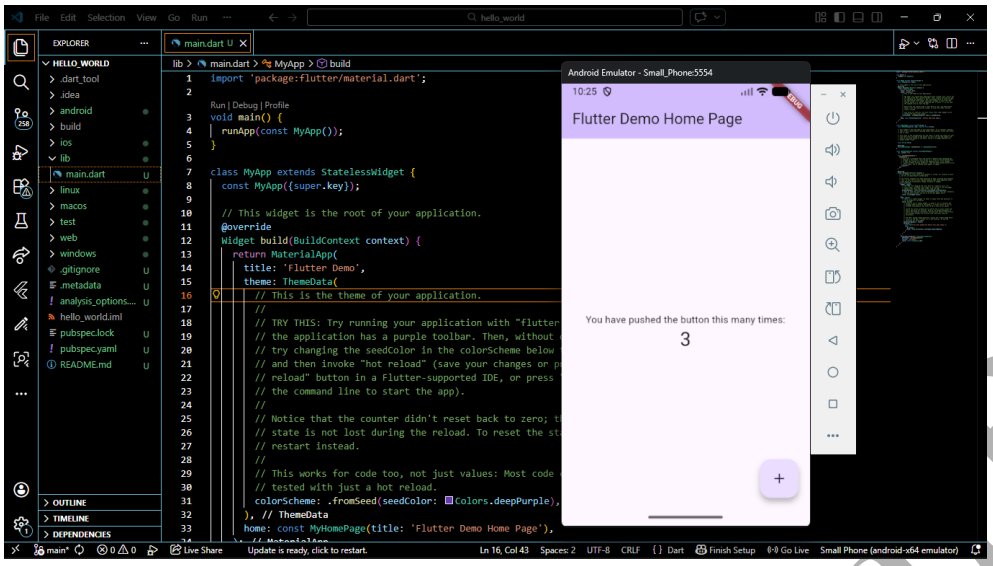
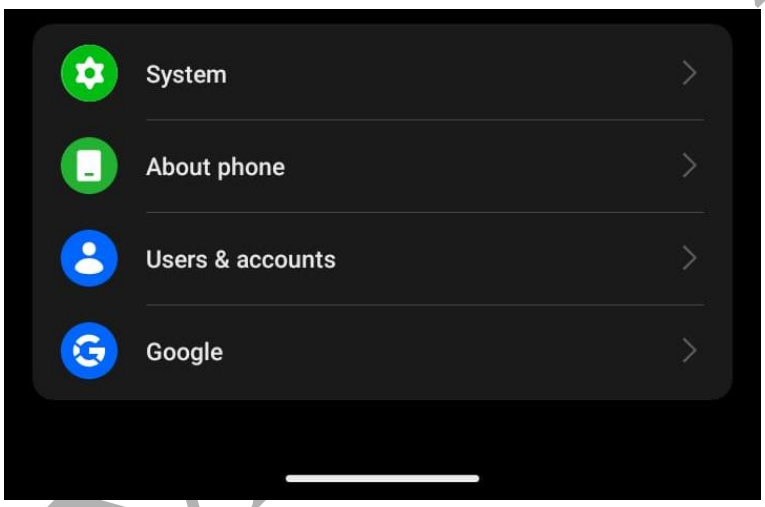


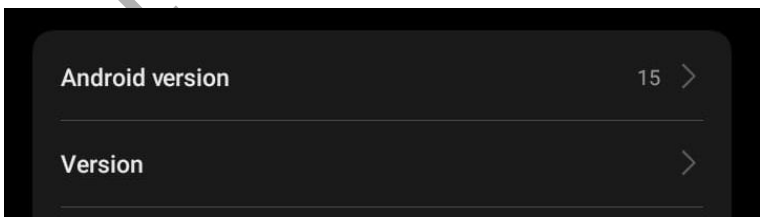
Praktikum 2: Menghubungkan Perangkat Android atau Emulator

Melanjutkan dari praktikum 1, Anda diminta untuk menjalankan aplikasi ke perangkat fisik (device Android atau iOS).

Cari About Phone di Pengatuhan device android



Klik Version



Klik 7 kali secara cepat di Build Number

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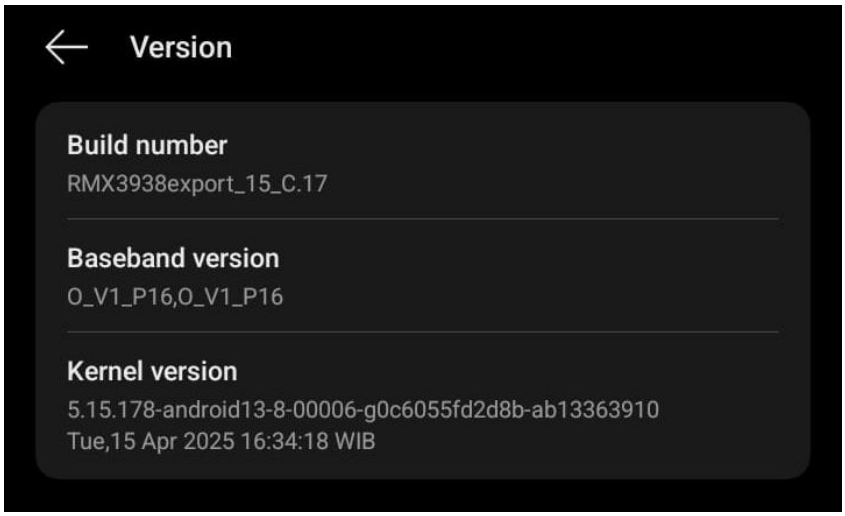
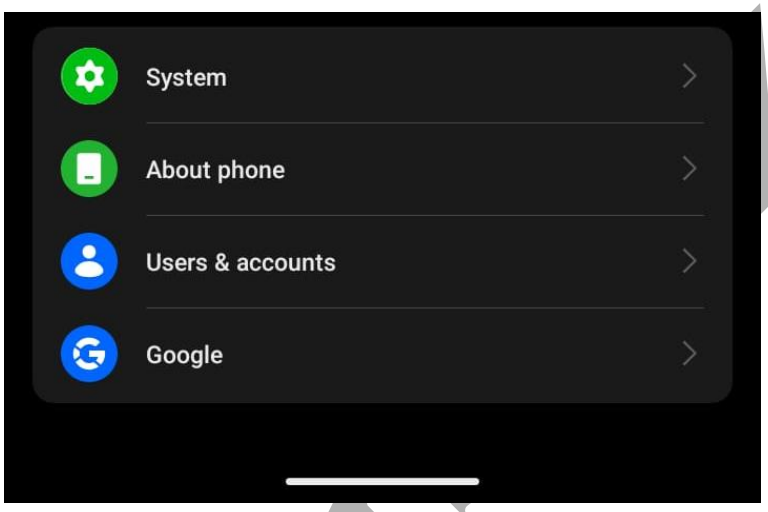
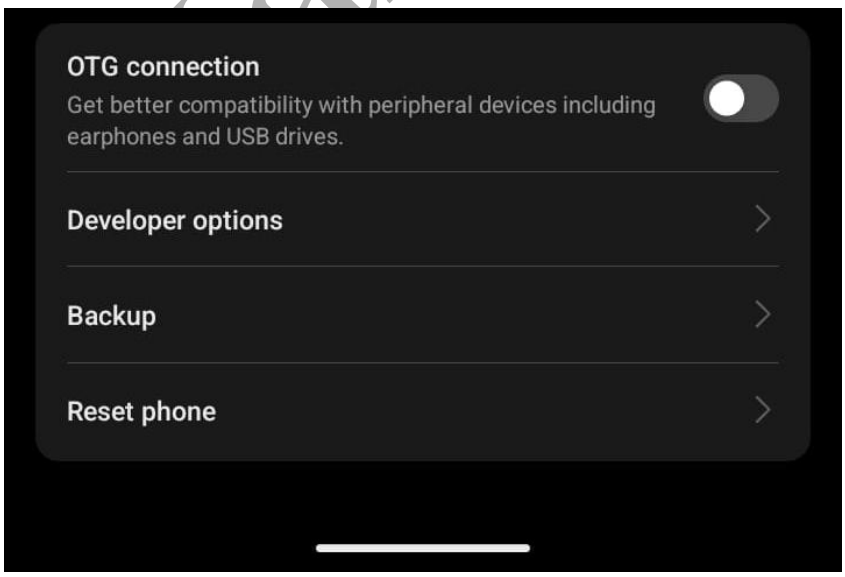


Pilih System



Pilih Developer options



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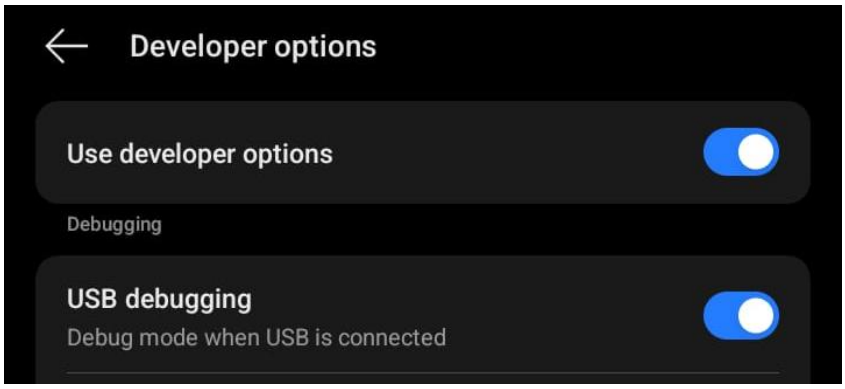
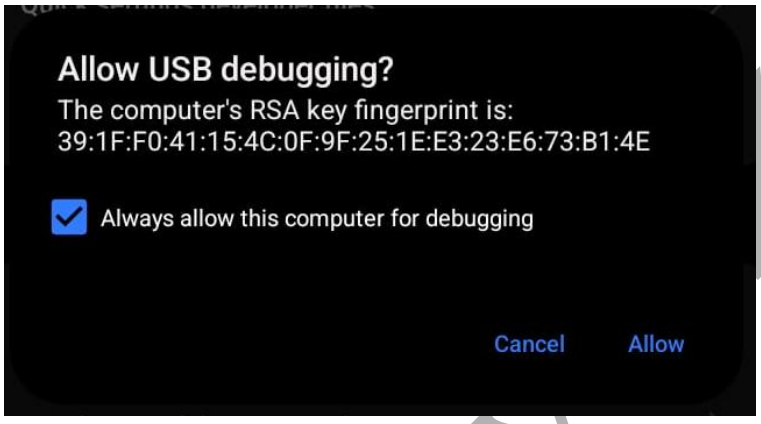
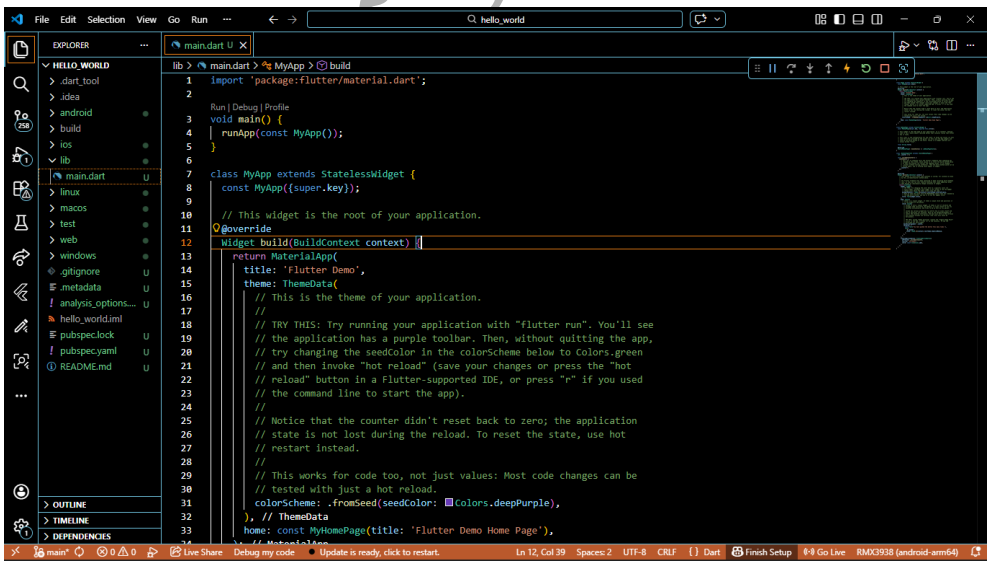
Klik USB debugging



Centang Always Allow dan klik Allow



Jalankan main.dart dengan device anddroid yg sudah si sambungkan dengan kabel USB ke laptop



Maka di device android akan muncul seperti dibawah

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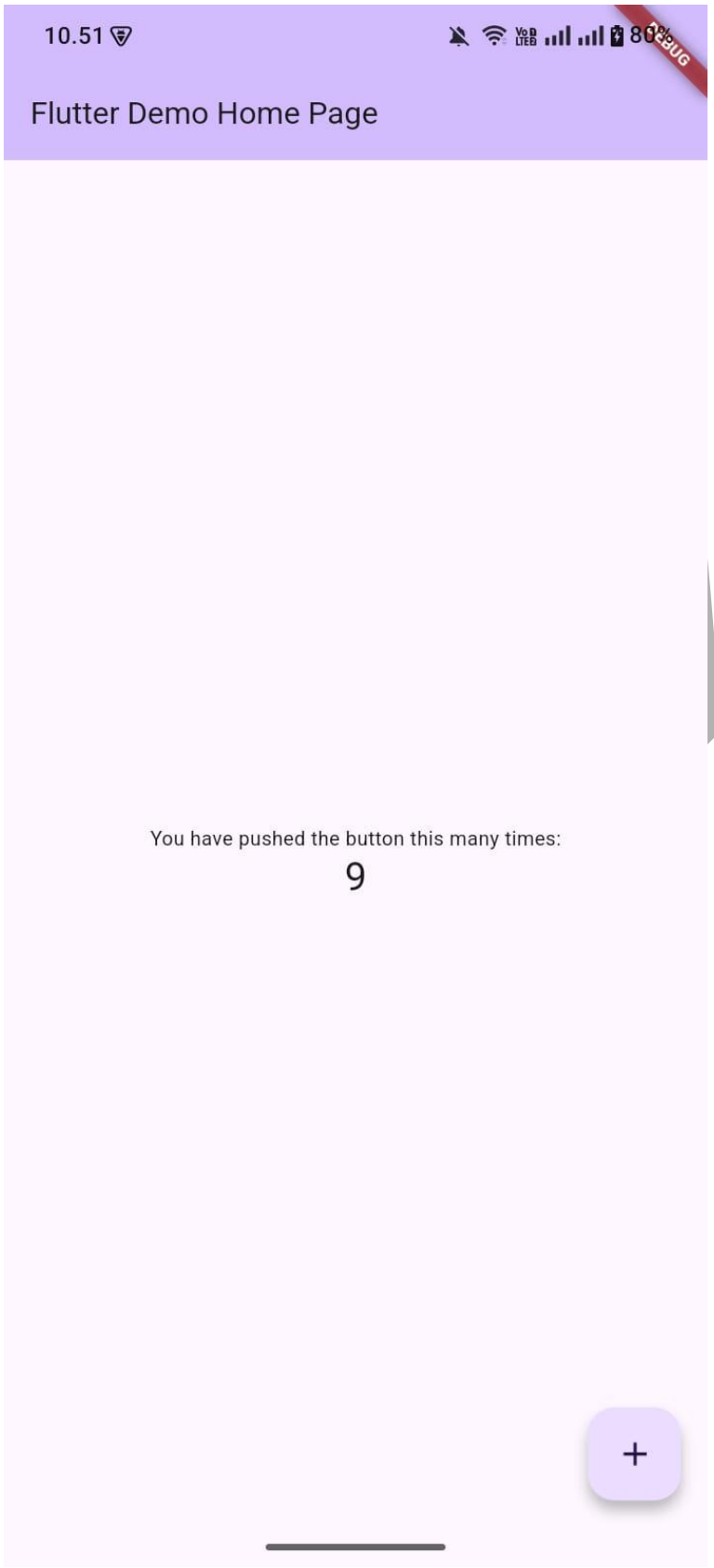

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Praktikum 4: Menerapkan Widget Dasar

Langkah 1: Text Widget

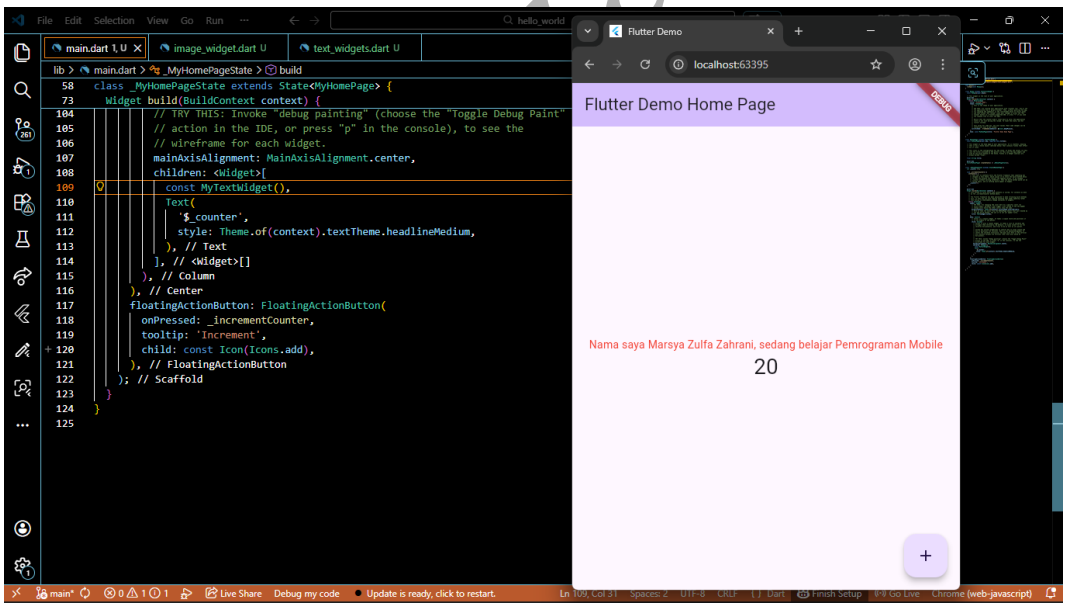
Buat folder baru basic_widgets di dalam folder lib. Kemudian buat file baru di dalam basic_widgets dengan nama text_widget.dart. Ketik atau salin kode program berikut ke project hello_world Anda pada file text_widget.dart.

```dart
import 'package:flutter/material.dart';

class MyTextWidget extends StatelessWidget {
  const MyTextWidget({Key? key}) : super(key: key);

  @override
  Widget build(BuildContext context) {
    return const Text(
      "Nama saya Marsya Zulfa Zahrani, sedang belajar Pemrograman Mobile",
      style: TextStyle(color: Colors.red, fontSize: 14),
      textAlign: TextAlign.center,
    ); // Text
  }
}
```

Lakukan import file text_widget.dart ke main.dart, lalu ganti bagian text widget dengan kode di atas. Maka hasilnya seperti gambar berikut.



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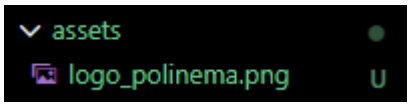
Langkah 2: Image Widget

Buat sebuah file image_widget.dart di dalam folder basic_widgets dengan isi kode berikut

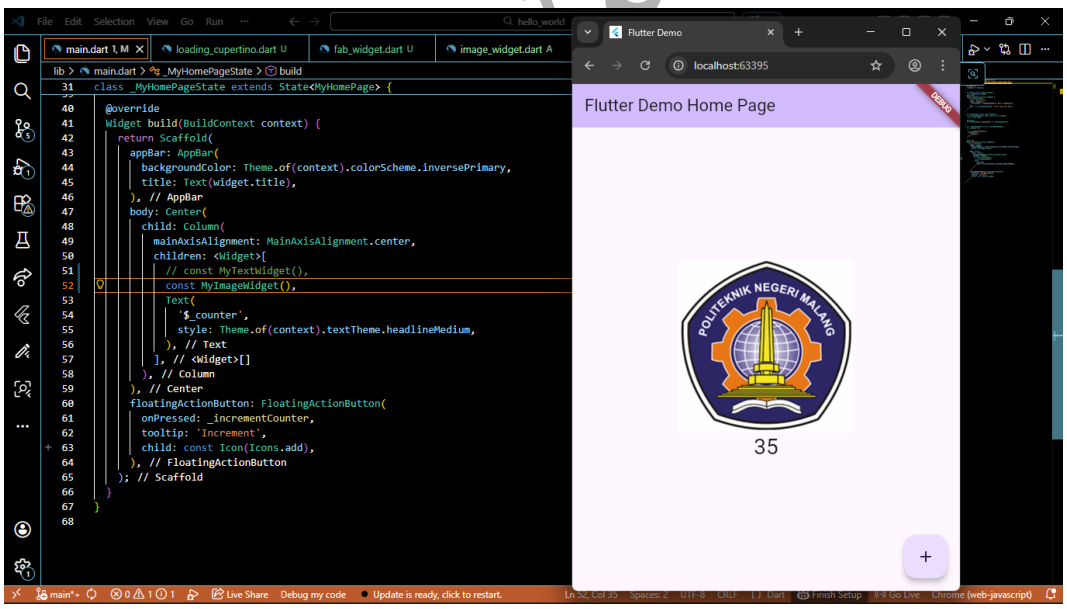
lib > basic_widgets > image_widget.dart > ...

```dart
import 'package:flutter/material.dart';

class MyImageWidget extends StatelessWidget {
  const MyImageWidget({Key? key}) : super(key: key);

  @override
  Widget build(BuildContext context) {
    return const Image(image: AssetImage("logo_polinema.png")
    ); // Image
  }
}
```

Lakukan penyesuaian asset pada file pubspec.yaml dan tambahkan file logo Anda di folder assets project hello_world.



Jangan lupa sesuaikan kode dan import di file main.dart kemudian akan tampil gambar seperti berikut.



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Praktikum 5: Menerapkan Widget Material Design dan iOS Cupertino

Langkah 1: Cupertino Button dan Loading Bar

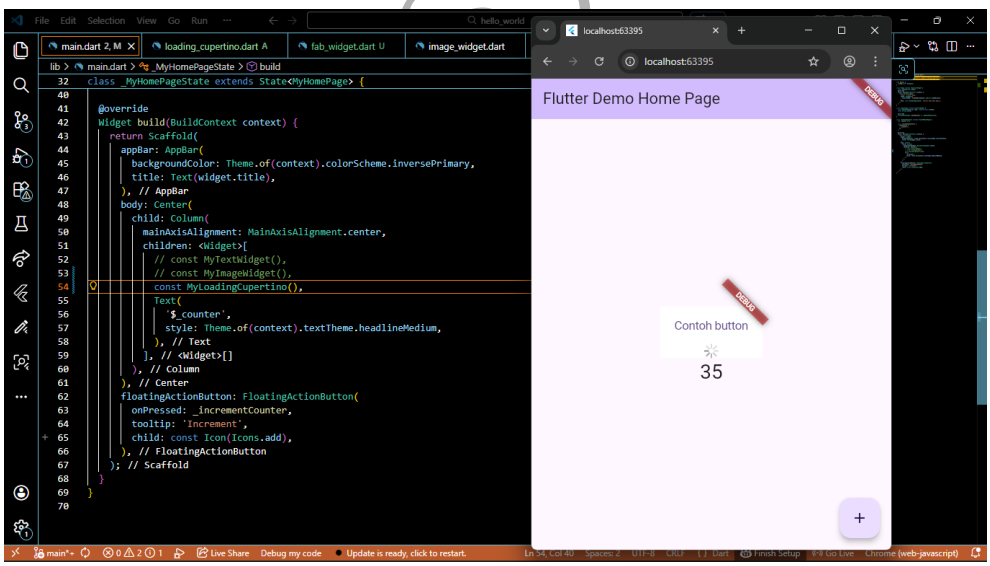
Buat file di basic_widgets > loading_cupertino.dart. Import stateless widget dari material dan cupertino. Lalu isi kode di dalam method Widget build adalah sebagai berikut.

lib > basic_widgets > loading_cupertino.dart > ...

```dart
import 'package:flutter/material.dart';
import 'package:flutter/cupertino.dart';

class MyLoadingCupertino extends StatelessWidget {
  const MyLoadingCupertino({Key? key}) : super(key: key);

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Container(
        margin: const EdgeInsets.only(top: 30),
        color: Colors.white,
        child: Column(
          children: <Widget>[
            CupertinoButton(
              child: const Text("Contoh button"),
              onPressed: () {},
            ), // CupertinoButton
            const CupertinoActivityIndicator(),
          ], // <Widget>[]
        ), // Column
      ), // Container
    ); // MaterialApp
  }
}
```



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Langkah 2: Floating Action Button (FAB)

Button widget terdapat beberapa macam pada flutter yaitu ButtonBar, DropdownButton, TextButton, FloatingActionButton, IconButton, OutlineButton, PopupMenuButton, dan ElevatedButton.

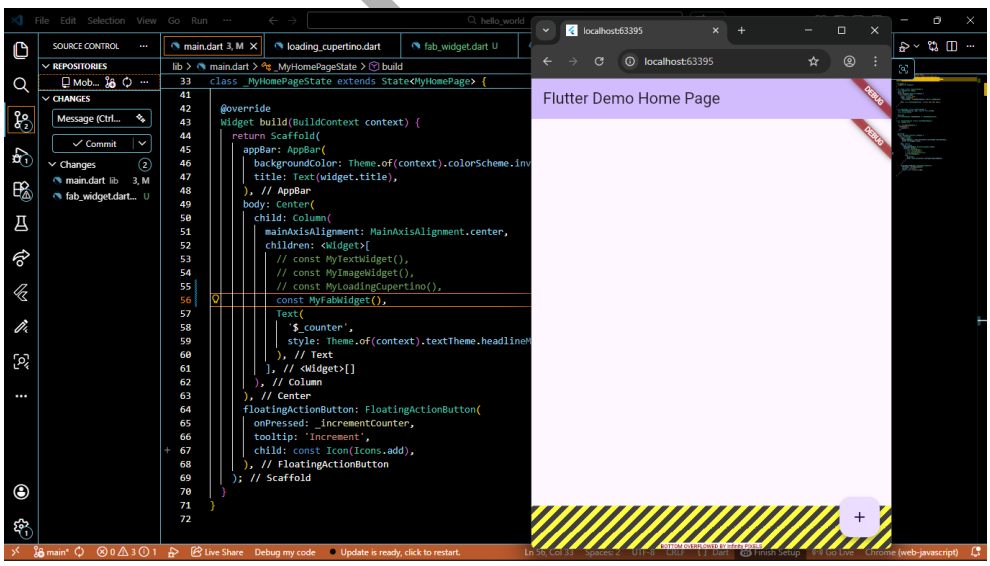
Buat file di basic_widgets > fab_widget.dart. Import stateless widget dari material. Lalu isi kode di dalam method Widget build adalah sebagai berikut.

lib > basic_widgets > fab_widget.dart > ...

```dart
import 'package:flutter/material.dart';

class MyFabWidget extends StatelessWidget {
  const MyFabWidget({Key? key}) : super(key: key);

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        floatingActionButton: FloatingActionButton(
          onPressed: () {
            // Add your onPressed code here!
          },
          backgroundColor: Colors.pink,
          child: const Icon(Icons.thumb_up),
        ), // FloatingActionButton
      ), // Scaffold
    ); // MaterialApp
  }
}
```



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Langkah 3: Scaffold Widget

Scaffold widget digunakan untuk mengatur tata letak sesuai dengan material design. Ubah isi kode main.dart seperti berikut.

main.dart

```dart
import 'package:flutter/material.dart';

void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({Key? key}) : super(key: key);

  // This widget is the root of your application.
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Flutter Demo',
      theme: ThemeData(
        primarySwatch: Colors.red,
      ),
      home: const MyHomePage(title: 'My Increment App'),
    );
  }
}

class MyHomePage extends StatefulWidget {
  const MyHomePage({Key? key, required this.title}) : super(key: key);

  final String title;

  @override
  State<MyHomePage> createState() => _MyHomePageState();
}

class _MyHomePageState extends State<MyHomePage> {
  int _counter = 0;

  void _incrementCounter() {
    setState(() {
      _counter++;
    });
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: Text(widget.title),
      ),
      body: Center(
        child: Column(
          mainAxisAlignment: MainAxisAlignment.center,
          children: <Widget>[
            const Text(
              'You have pushed the button this many times:',
            ),
            Text(
              '$_counter',
              style: Theme.of(context).textTheme.headlineMedium,
            ),
          ],
        ),
      ),
      bottomNavigationBar: BottomAppBar(
        child: Container(
          height: 50.0,
        ),
      ),
      floatingActionButton: FloatingActionButton(
        onPressed: _incrementCounter,
        tooltip: 'Increment Counter',
        child: const Icon(Icons.add),
      ),
      floatingActionButtonLocation: FloatingActionButtonLocation.centerDocked,
    );
  }
}
```

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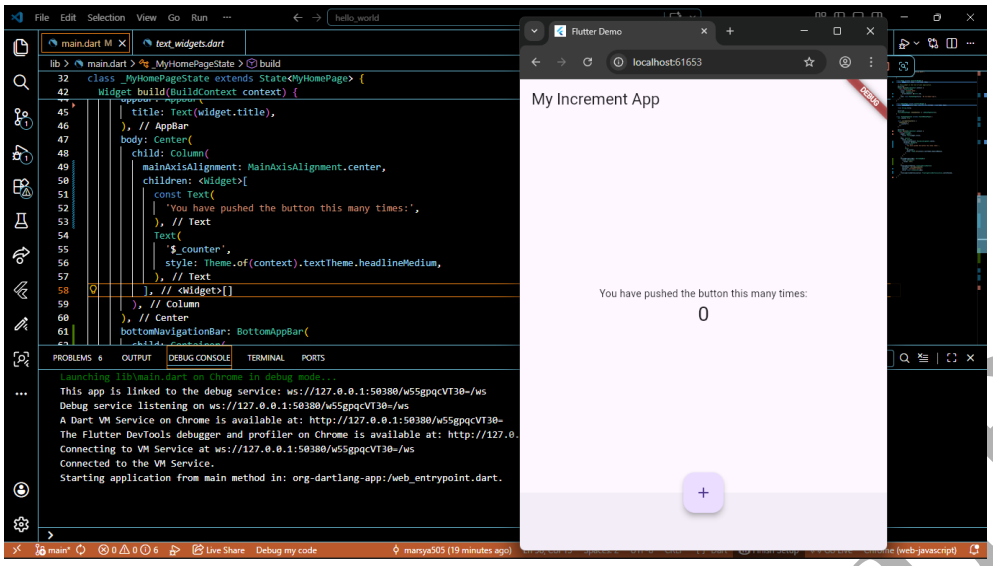

Langkah 4: Dialog Widget

Dialog widget pada flutter memiliki dua jenis dialog yaitu AlertDialog dan SimpleDialog. Ubah isi kode main.dart seperti berikut.

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main.dart

```dart
import 'package:flutter/material.dart';

void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({Key? key}) : super(key: key);

  @override
  Widget build(BuildContext context) {
    return const MaterialApp(
      home: Scaffold(
        body: MyLayout(),
      ),
    );
  }
}

class MyLayout extends StatelessWidget {
  const MyLayout({Key? key}) : super(key: key);

  @override
  Widget build(BuildContext context) {
    return Padding(
      padding: const EdgeInsets.all(8.0),
      child: ElevatedButton(
        child: const Text('Show alert'),
        onPressed: () {
          showAlertDialog(context);
        },
      ),
    );
  }
}

showAlertDialog(BuildContext context) {
  // set up the button
  Widget okButton = TextButton(
    child: const Text("OK"),
    onPressed: () {
      Navigator.pop(context);
    },
  );

  // set up the AlertDialog
  AlertDialog alert = AlertDialog(
    title: const Text("My title"),
    content: const Text("This is my message."),
    actions: [
      okButton,
    ],
  );

  // show the dialog
  showDialog(
    context: context,
    builder: (BuildContext context) {
      return alert;
    },
  );
}

class MyHomePage extends StatefulWidget {
  const MyHomePage({Key? key, required this.title}) : super(key: key);

  final String title;

  @override
  State<MyHomePage> createState() => _MyHomePageState();
}

class _MyHomePageState extends State<MyHomePage> {
  int _counter = 0;

  void _incrementCounter() {
    setState(() {
      _counter++;
    });
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: Text(widget.title),
      ),
      body: Center(
        child: Column(
          mainAxisAlignment: MainAxisAlignment.center,
          children: <Widget>[
            const Text(
              'You have pushed the button this many times:',
            ),
            Text(
              '$_counter',
              style: Theme.of(context).textTheme.headlineMedium,
            ),
          ],
        ),
      ),
      bottomNavigationBar: BottomAppBar(
        child: Container(
          height: 50.0,
        ),
      ),
      floatingActionButton: FloatingActionButton(
        onPressed: _incrementCounter,
        tooltip: 'Increment Counter',
        child: const Icon(Icons.add),
      ),
      floatingActionButtonLocation: FloatingActionButtonLocation.centerDocked,
    );
  }
}
```

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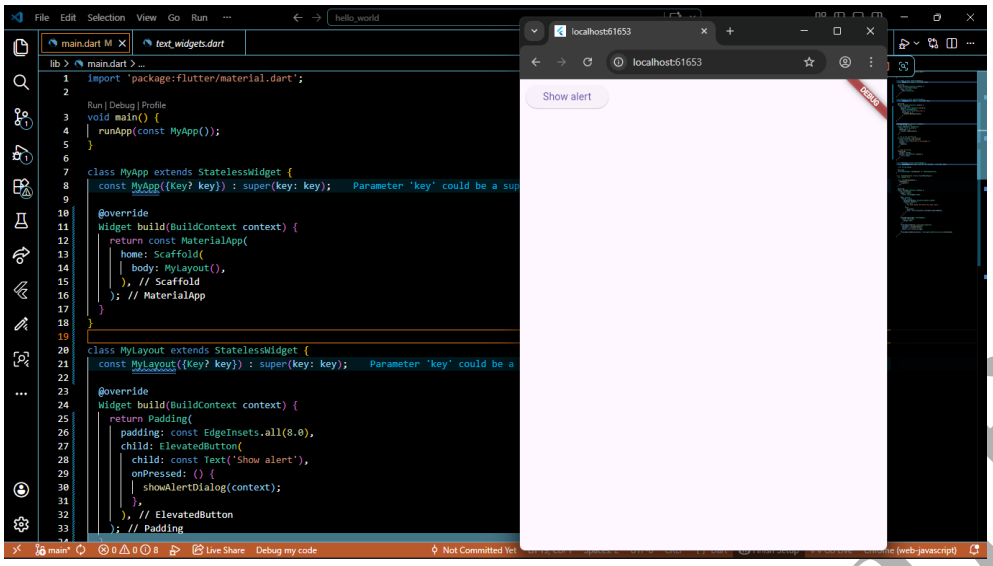

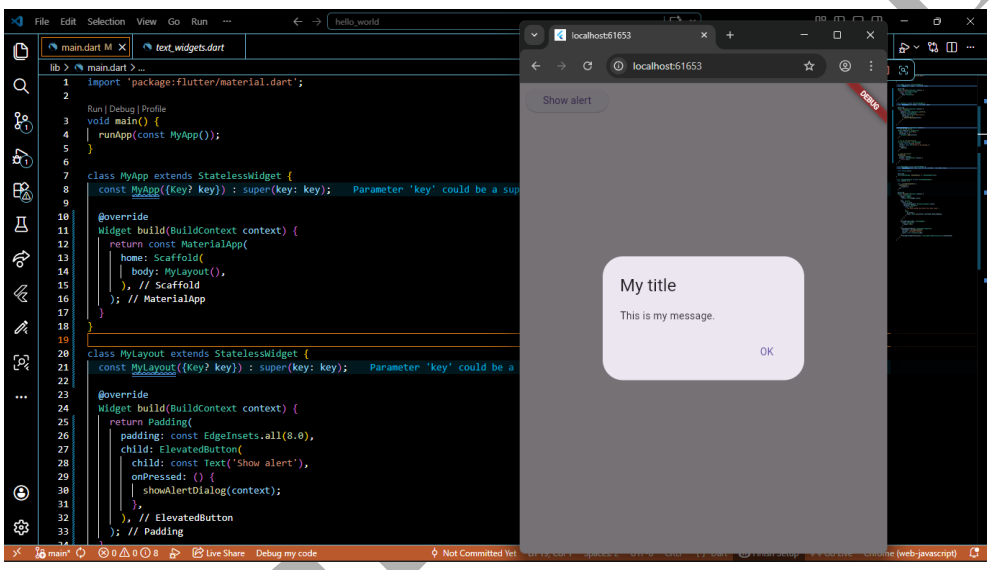

Langkah 5: Input dan Selection Widget

Flutter menyediakan widget yang dapat menerima input dari pengguna aplikasi yaitu antara lain Checkbox, Date and Time Pickers, Radio Button, Slider, Switch, TextField.

Contoh penggunaan TextField widget adalah sebagai berikut:

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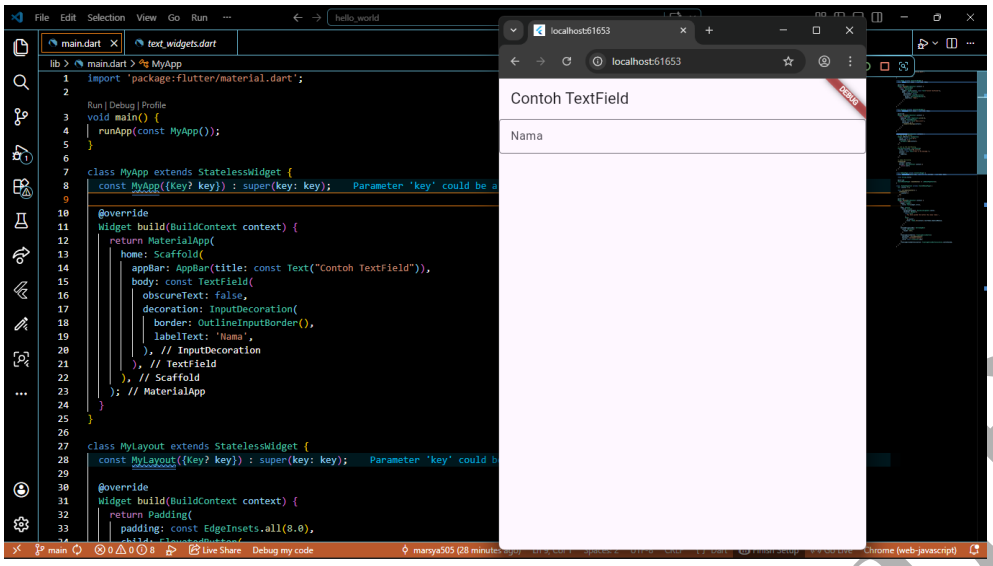

Langkah 6: Date and Time Pickers

Date and Time Pickers termasuk pada kategori input dan selection widget, berikut adalah contoh penggunaan Date and Time Pickers.

Marsya Zulfa

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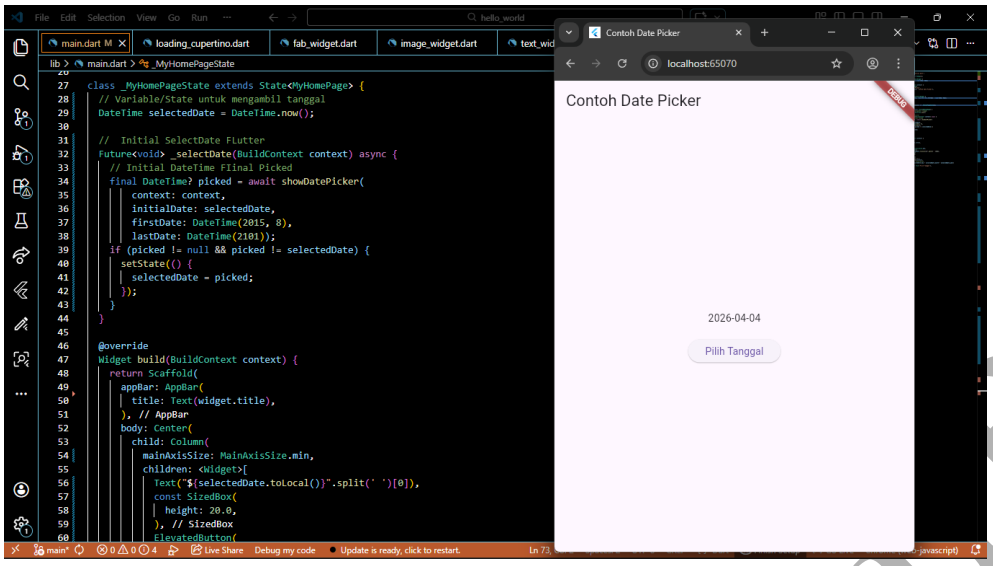
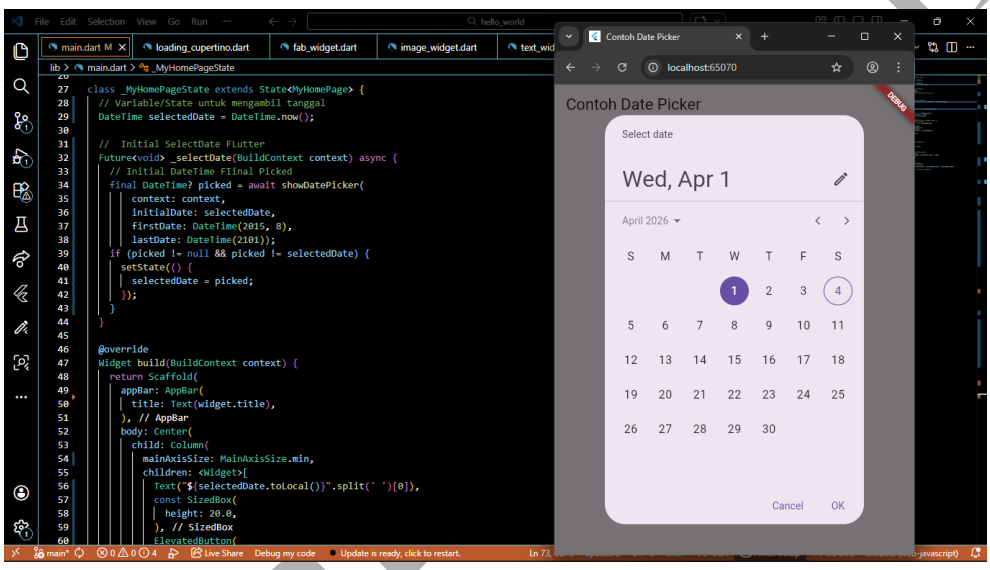
main.dart

```dart
import 'dart:async';
import 'package:flutter/material.dart';

void main() => runApp(const MyApp());

class MyApp extends StatelessWidget {
  const MyApp({Key? key}) : super(key: key);

  @override
  Widget build(BuildContext context) {
    return const MaterialApp(
      title: 'Contoh Date Picker',
      home: MyHomePage(title: 'Contoh Date Picker'),
    );
  }
}

class MyHomePage extends StatefulWidget {
  const MyHomePage({Key? key, required this.title}) : super(key: key);

  final String title;

  @override
  _MyHomePageState createState() => _MyHomePageState();
}

class _MyHomePageState extends State<MyHomePage> {
  // Variable/State untuk mengambil tanggal
  DateTime selectedDate = DateTime.now();

  //  Initial SelectDate FLutter
  Future<void> _selectDate(BuildContext context) async {
    // Initial DateTime FIinal Picked
    final DateTime? picked = await showDatePicker(
        context: context,
        initialDate: selectedDate,
        firstDate: DateTime(2015, 8),
        lastDate: DateTime(2101));
    if (picked != null && picked != selectedDate) {
      setState(() {
        selectedDate = picked;
      });
    }
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: Text(widget.title),
      ),
      body: Center(
        child: Column(
          mainAxisSize: MainAxisSize.min,
          children: <Widget>[
            Text("${selectedDate.toLocal()}".split(' ')[0]),
            const SizedBox(
              height: 20.0,
            ),
            ElevatedButton(
              onPressed: () => {
                _selectDate(context),
                // ignore: avoid_print
                print(selectedDate.day + selectedDate.month + selectedDate.year)
              },
              child: const Text('Pilih Tanggal'),
            ),
          ],
        ),
      ),
    );
  }
}
```

Zulfa

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ABSENT : 11
CLASS : SIB 2G
OBJECT : MOBILE PROGRAMMING